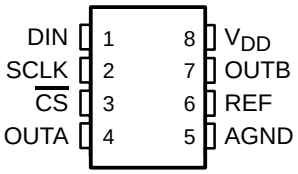
## features

- **Dual 10-Bit Voltage Output DAC**
- **Programmable Settling Time**
  - **3 μs in Fast Mode**
  - **10 μs in Slow Mode**
- **Compatible With TMS320 and SPI™ Serial Ports**
- **Differential Nonlinearity <0.1 LSB Typ**
- **Monotonic Over Temperature**
- **Direct Replacement for TLC5617A**

## applications

- **Digital Servo Control Loops**
- **Digital Offset and Gain Adjustment**
- **Industrial Process Control**
- **Machine and Motion Control Devices**
- **Mass Storage Devices**

**D PACKAGE**
**(TOP VIEW)**




## description

The TLV5617A is a dual 10-bit voltage output DAC with a flexible 3-wire serial interface. The serial interface is compatible with TMS320, SPI™, QSPI™, and Microwire™ serial ports. It is programmed with a 16-bit serial string containing 4 control bits and 10 data bits.

The resistor string output voltage is buffered by a x2 gain rail-to-rail output buffer. The buffer features a Class-AB output stage to improve stability and reduce settling time. The programmable settling time of the DAC allows the designer to optimize speed versus power dissipation.

Implemented with a CMOS process, the device is designed for single supply operation from 2.7 V to 5.5 V. It is available in an 8-pin SOIC package in standard commercial and industrial temperature ranges.

**AVAILABLE OPTIONS**

| $T_A$ | PACKAGE |
|---|---|
| | SOIC (D) |
| 0°C to 70°C | TLV5617ACD |
| –40°C to 85°C | TLV5617AID |

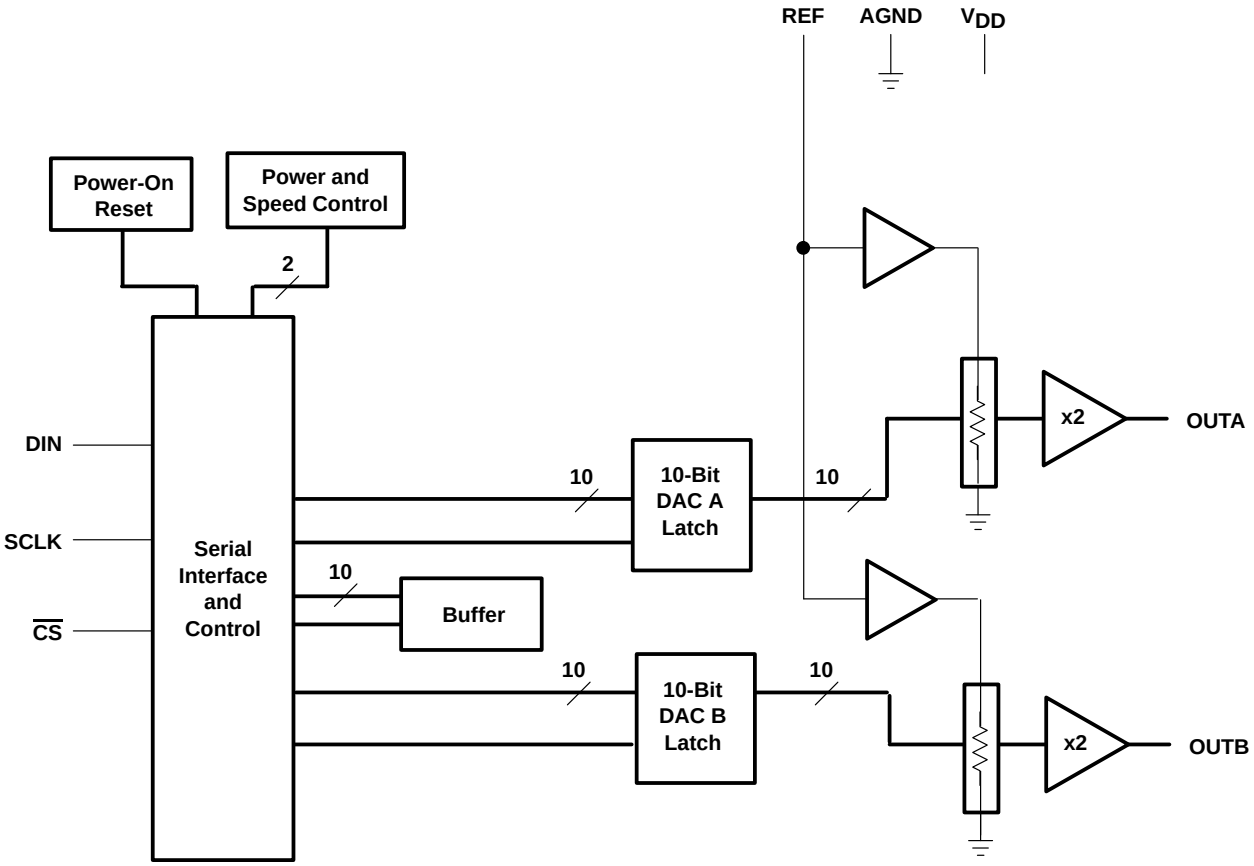
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## functional block diagram




### Terminal Functions

| TERMINAL NAME | TERMINAL NO. | I/O/P | DESCRIPTION |
|---|---|---|---|
| AGND | 5 | P | Ground |
| $\overline{CS}$ | 3 | I | Chip select. Digital input active low, used to enable/disable inputs. |
| DIN | 1 | I | Digital serial data input |
| OUTA | 4 | O | DAC A analog voltage output |
| OUTB | 7 | O | DAC B analog voltage output |
| REF | 6 | I | Analog reference voltage input |
| SCLK | 2 | I | Digital serial clock input |
| $V_{DD}$ | 8 | P | Positive power supply |

### absolute maximum ratings over operating free-air temperature range (unless otherwise noted)†

Supply voltage ($V_{DD}$ to AGND) . . . . . 7 V
Reference input voltage range . . . . . – 0.3 V to $V_{DD}$ + 0.3 V
Digital input voltage range . . . . . – 0.3 V to $V_{DD}$ + 0.3 V
Operating free-air temperature range, $T_A$: TLV5617AC . . . . . 0°C to 70°C
TLV5617AI . . . . . –40°C to 85°C
Storage temperature range, $T_{stg}$ . . . . . –65°C to 150°C
Lead temperature 1,6 mm (1/16 inch) from case for 10 seconds . . . . . 260°C

† Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

### recommended operating conditions

| | | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|
| Supply voltage, $V_{DD}$ | $V_{DD}$ = 5 V | 4.5 | 5 | 5.5 | V |
| | $V_{DD}$ = 3 V | 2.7 | 3 | 3.3 | |
| Power on reset, POR | | 0.55 | | 2 | V |
| High-level digital input voltage, $V_{IH}$ | $V_{DD}$ = 2.7 V | 2 | | | V |
| | $V_{DD}$ = 5.5 V | 2.4 | | | |
| Low-level digital input voltage, $V_{IL}$ | $V_{DD}$ = 2.7 V | | | 0.6 | V |
| | $V_{DD}$ = 5.5 V | | | 1 | |
| Reference voltage, $V_{ref}$ to REF terminal | $V_{DD}$ = 5 V (see Note 1) | AGND | 2.048 | $V_{DD}$–1.5 | V |
| | $V_{DD}$ = 3 V (see Note 1) | AGND | 1.024 | $V_{DD}$–1.5 | |
| Load resistance, $R_L$ | | 2 | | | kΩ |
| Load capacitance, $C_L$ | | | | 100 | pF |
| Clock frequency, $f_{CLK}$ | | | | 20 | MHz |
| Operating free-air temperature, $T_A$ | TLV5617AC | 0 | | 70 | °C |
| | TLV5617AI | –40 | | 85 | |

NOTE 1: Due to the x2 output buffer, a reference input voltage ≥ ($V_{DD}$–0.4 V)/2 causes clipping of the transfer function.

## electrical characteristics over recommended operating conditions (unless otherwise noted)

### power supply

| PARAMETER | | TEST CONDITIONS | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $I_{DD}$ | Power supply current | No load, All inputs = AGND or $V_{DD}$, DAC latch = 0x800 | Fast | | 1.6 | 2.5 | mA |
| | | | Slow | | 0.6 | 1 | |
| | Power down supply current | | | | 1 | | μA |
| PSRR | Power supply rejection ratio | Zero scale, See Note 2 | | | –65 | | dB |
| | | Full scale, See Note 3 | | | –65 | | |

NOTES: 2. Power supply rejection ratio at zero scale is measured by varying $V_{DD}$ and is given by:
$PSRR = 20 \log [(E_{ZS}(V_{DD}max) – E_{ZS}(V_{DD}min)/V_{DD}max]$
3. Power supply rejection ratio at full scale is measured by varying $V_{DD}$ and is given by:
$PSRR = 20 \log [(E_G(V_{DD}max) – E_G(V_{DD}min)/V_{DD}max]$

### static DAC specifications

| PARAMETER | | TEST CONDITIONS | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| | Resolution | | | 10 | | | bits |
| INL | Integral nonlinearity | See Note 4 | | | ±0.7 | ±1 | LSB |
| DNL | Differential nonlinearity | See Note 5 | | | ±0.1 | ±0.5 | LSB |
| $E_{ZS}$ | Zero-scale error (offset error at zero scale) | See Note 6 | | | | ±12 | mV |
| $E_{ZS}$ TC | Zero-scale-error temperature coefficient | See Note 7 | | | 3 | | ppm/°C |
| $E_G$ | Gain error | See Note 8 | $V_{DD}$ = 2.7 V to 3.3 V | | | ±0.6 | % full scale V |
| | | | $V_{DD}$ = 4.5 V to 5.5 V | | | ±0.29 | |
| $E_G$ $T_C$ | Gain-error temperature coefficient | See Note 9 | | | 1 | | ppm/°C |

NOTES: 4. The relative accuracy of integral nonlinearity (INL), sometimes referred to as linearity error, is the maximum deviation of the output from the line between zero and full scale, excluding the effects of zero-code and full-scale errors.
5. The differential nonlinearity (DNL), sometimes referred to as differential error, is the difference between the measured and ideal 1-LSB amplitude change of any two adjacent codes.
6. Zero-scale error is the deviation from zero voltage output when the digital input code is zero.
7. Zero-scale-error temperature coefficient is given by: $E_{ZS}$ TC = $[E_{ZS}(T_{max}) – E_{ZS}(T_{min})]/2V_{ref} \times 10^6/(T_{max} – T_{min})$.
8. Gain error is the deviation from the ideal output ($2V_{ref}$ – 1 LSB) with an output load of 10 kΩ.
9. Gain temperature coefficient is given by: $E_G$ $T_C$ = $[E_G(T_{max}) – E_g(T_{min})]/2V_{ref} \times 10^6/(T_{max} – T_{min})$.

### output specifications

| PARAMETER | | TEST CONDITIONS | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|
| $V_O$ | Output voltage range | $R_L$ = 10 kΩ | | | $V_{DD}$–0.4 | V |
| | Output load regulation accuracy | $V_O$ = 4.096 V, 2.048 V, $R_L$ = 2 kΩ to 10 kΩ | | ±0.1 | | % FS |

### reference input

| PARAMETER | | TEST CONDITIONS | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $V_I$ | Input voltage range | | | 0 | | $V_{DD}$–1.5 | V |
| $R_I$ | Input resistance | | | | 10 | | MΩ |
| $C_I$ | Input capacitance | | | | 5 | | pF |
| | Reference input bandwidth | REF = 0.2 $V_{pp}$ + 1.024 V dc | Fast | | 1.3 | | MHz |
| | | | Slow | | 525 | | kHz |
| | Reference feedthrough | REF = 1 $V_{pp}$ at 1 kHz + 1.024 V dc (see Note 10) | | | –80 | | dB |

NOTE 10: Reference feedthrough is measured at the DAC output with an input code = 0x000.

## electrical characteristics over recommended operating conditions (unless otherwise noted) (Continued)

### digital inputs

| PARAMETER | | TEST CONDITIONS | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|
| $I_{IH}$ | High-level digital input current | $V_I = V_{DD}$ | | | 1 | µA |
| $I_{IL}$ | Low-level digital input current | $V_I = 0$ V | –1 | | | µA |
| $C_i$ | Input capacitance | | | 8 | | pF |

### analog output dynamic performance

| PARAMETER | | TEST CONDITIONS | | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| $t_{s(FS)}$ | Output settling time, full scale | $R_L$ = 10 kΩ, $C_L$ = 100 pF, See Note 11 | Fast | | 1 | 3 | µs |
| | | | Slow | | 3 | 10 | |
| $t_{s(CC)}$ | Output settling time, code to code | $R_L$ = 10 kΩ, $C_L$ = 100 pF, See Note 12 | Fast | | 1 | | µs |
| | | | Slow | | 2 | | |
| SR | Slew rate | $R_L$ = 10 kΩ, $C_L$ = 100 pF, See Note 13 | Fast | | 3 | | V/µs |
| | | | Slow | | 0.5 | | |
| | Glitch energy | DIN = 0 to 1, FCLK = 100 kHz, $\overline{CS}$ = $V_{DD}$ | | | 5 | | nV–s |
| SNR | Signal-to-noise ratio | $f_S$ = 102 kSPS, $f_{out}$ = 1 kHz, $R_L$ = 10 kΩ, $C_L$ = 100 pF | | | 68 | | dB |
| SINAD | Signal-to-noise + distortion | | | | 65 | | |
| THD | Total harmonic distortion | | | | –62 | | |
| SFDR | Spurious free dynamic range | | | | 64 | | |

NOTES: 11. Settling time is the time for the output signal to remain within ±0.5 LSB of the final measured value for a digital input code change of 0x020 to 0xFDC and 0xFDC to 0x020 respectively. Not tested, assured by design.
12. Settling time is the time for the output signal to remain within ± 0.5 LSB of the final measured value for a digital input code change of one count. Not tested, assured by design.
13. Slew rate determines the time it takes for a change of the DAC output from 10% to 90% of full-scale voltage.

## digital input timing requirements

| | | | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|---|
| $t_{su(CS–CK)}$ | Setup time, $\overline{CS}$ low before first negative SCLK edge | $V_{DD}$ = 2.7 V to 3.3 V | 10 | | | ns |
| | | $V_{DD}$ = 4.5 V to 5.5 V | 5 | | | |
| $t_{su(C16-CS)}$ | Setup time, 16th negative SCLK edge before $\overline{CS}$ rising edge | | 10 | | | ns |
| $t_{wH}$ | SCLK pulse width high | | 25 | | | ns |
| $t_{wL}$ | SCLK pulse width low | | 25 | | | ns |
| $t_{su(D)}$ | Setup time, data ready before SCLK falling edge | $V_{DD}$ = 2.7 V to 3.3 V | 10 | | | ns |
| | | $V_{DD}$ = 4.5 V to 5.5 V | 5 | | | |
| $t_{h(D)}$ | Hold time, data held valid after SCLK falling edge | $V_{DD}$ = 2.7 V to 3.3 V | 10 | | | ns |
| | | $V_{DD}$ = 4.5 V to 5.5 V | 5 | | | |

## timing requirements

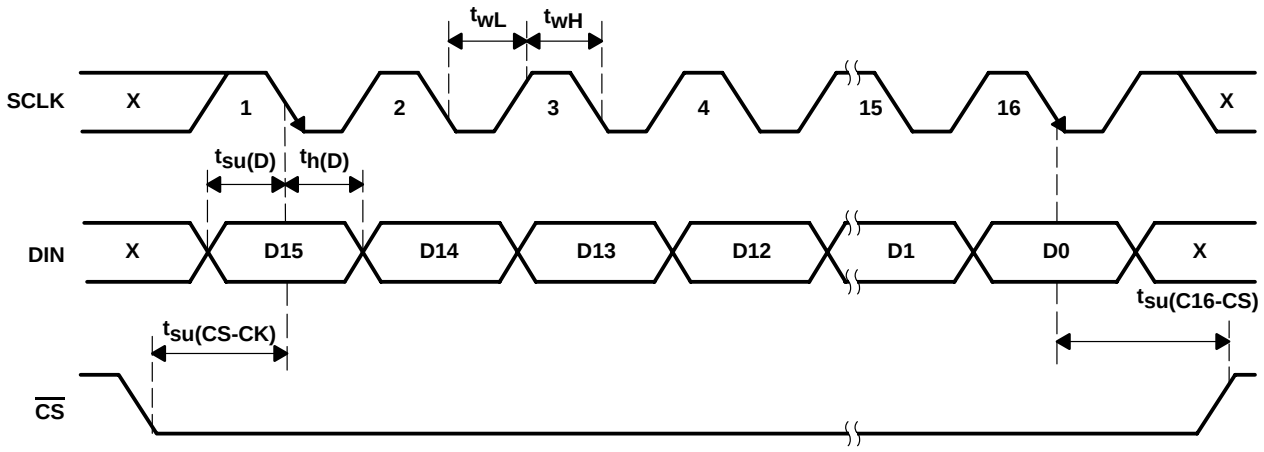



**Figure 1. Timing Diagram**

## TYPICAL CHARACTERISTICS

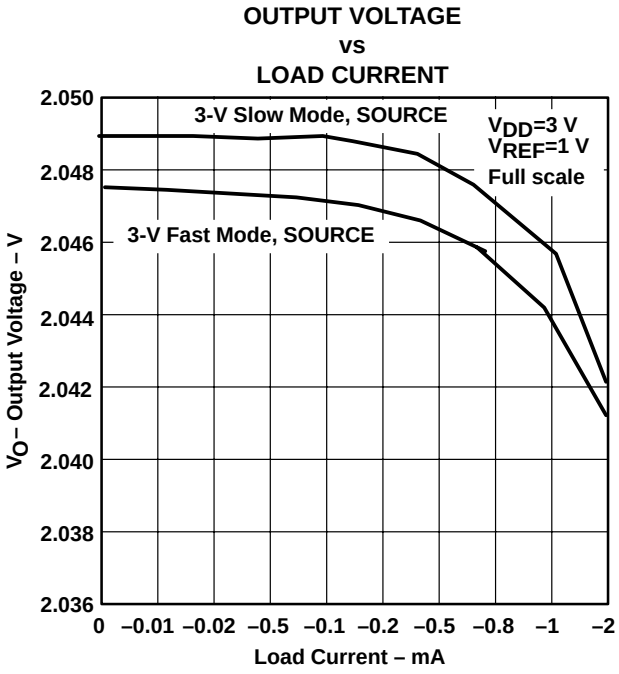



Figure 2

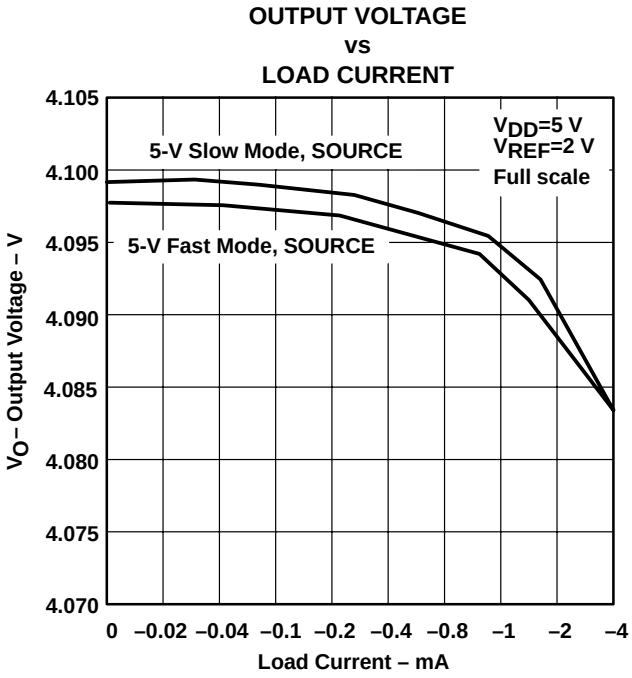



Figure 3

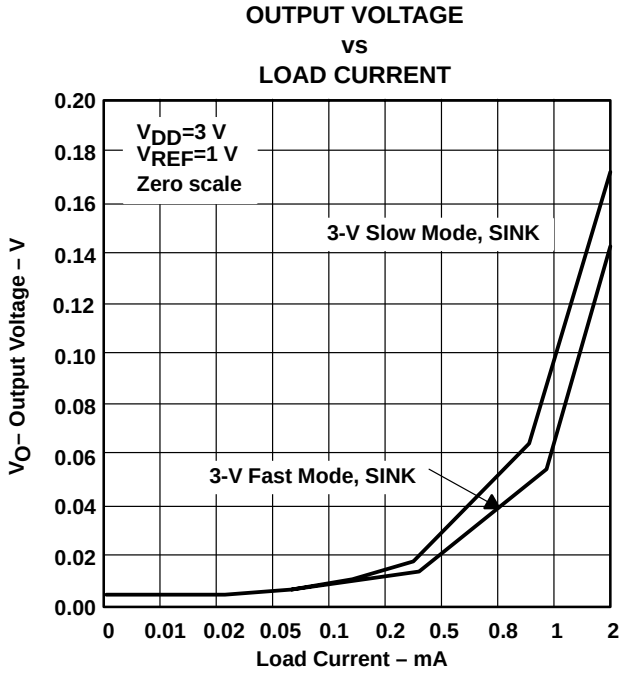



Figure 4

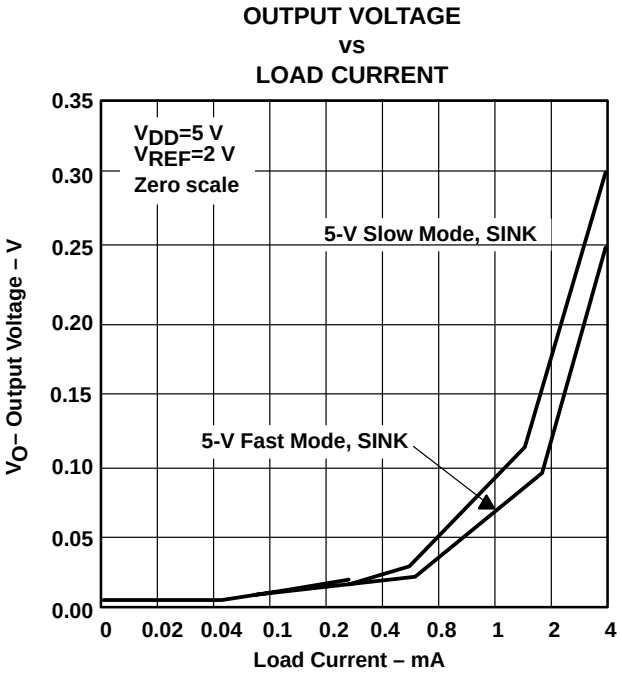



Figure 5

## TYPICAL CHARACTERISTICS

**SUPPLY CURRENT vs FREE-AIR TEMPERATURE**

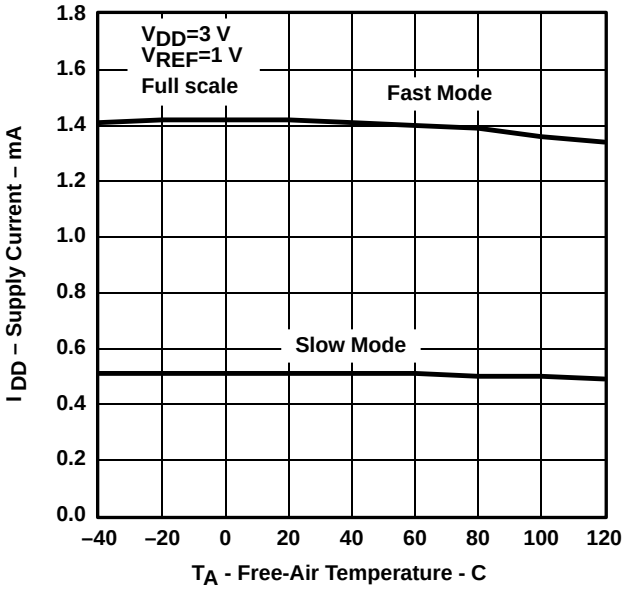



**Figure 6**

**SUPPLY CURRENT vs FREE-AIR TEMPERATURE**

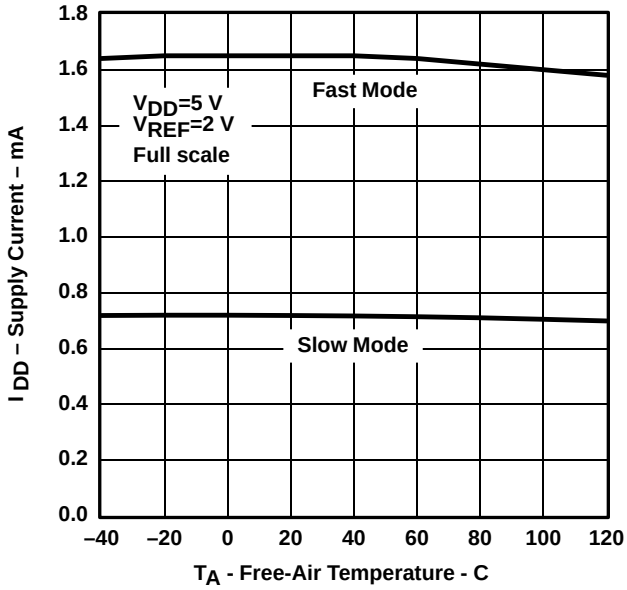



**Figure 7**

**TOTAL HARMONIC DISTORTION vs FREQUENCY**

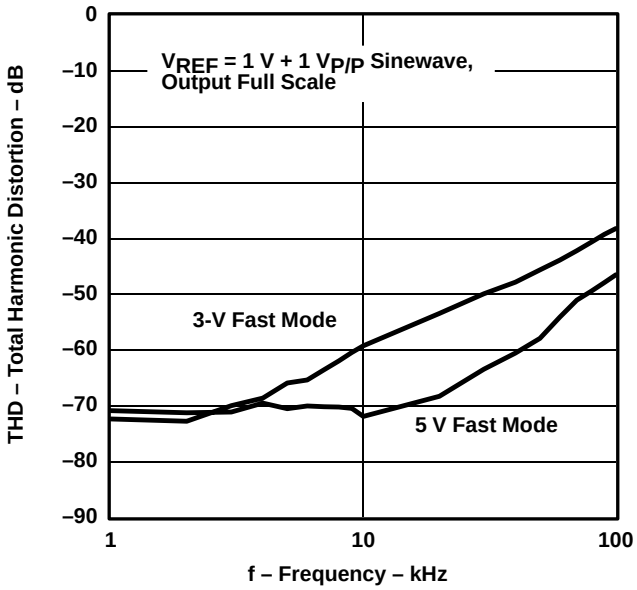



**Figure 8**

**TOTAL HARMONIC DISTORTION vs FREQUENCY**

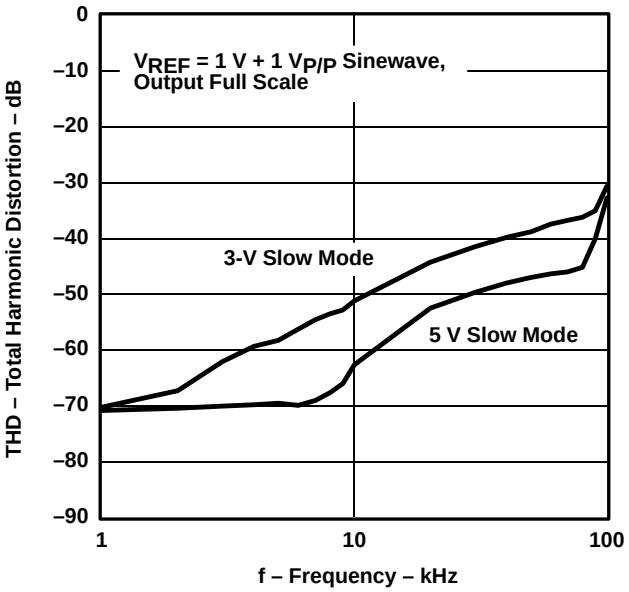



**Figure 9**

## TYPICAL CHARACTERISTICS

**INTEGRAL NONLINEARITY ERROR vs DIGITAL CODE**

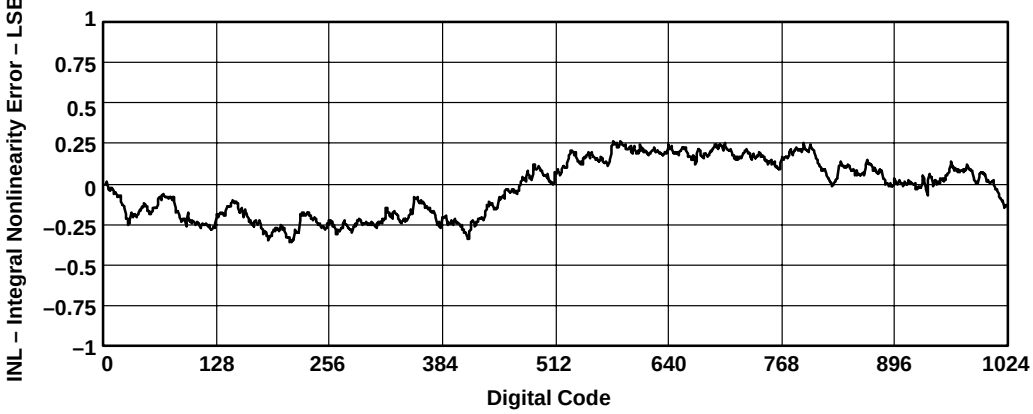



**Figure 10**

**DIFFERENTIAL NONLINEARITY ERROR vs DIGITAL CODE**

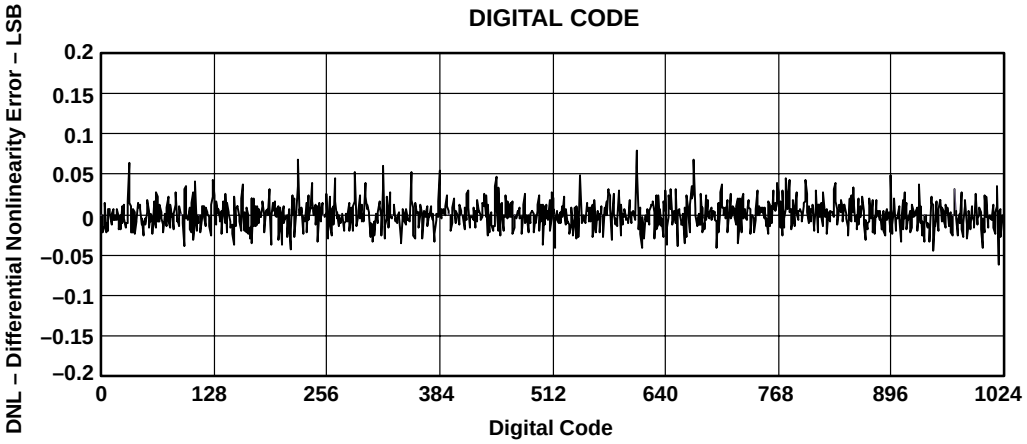



**Figure 11**

## APPLICATION INFORMATION

### general function

The TLV5617A is a dual 10-bit, single-supply DAC, based on a resistor-string architecture. It consists of a serial interface, speed and power-down control logic, a resistor string, and a rail-to-rail output buffer.

The output voltage (full scale determined by the reference) is given by:

$$2\ \mathrm{REF}\ \frac{\mathrm{CODE}}{2^n}\ [\mathrm{V}]$$

Where REF is the reference voltage and CODE is the digital input value within the range of $0_{10}$ to $2^n$–1, where n=10 (bits). The 16-bit data word, consisting of control bits and the new DAC value, is illustrated in the *data format* section. A power-on reset initially resets the internal latches to a defined state (all bits zero).

### serial interface

A falling edge of $\overline{CS}$ starts shifting the data bit-per-bit (starting with the MSB) to the internal register on the falling edges of SCLK. After 16 bits have been transferred or $\overline{CS}$ rises, the content of the shift register is moved to the target latches (DAC A, DAC B, BUFFER, CONTROL), depending on the control bits within the data word.

Figure 12 shows examples of how to connect the TLV5617A to TMS320, SPI, and Microwire.

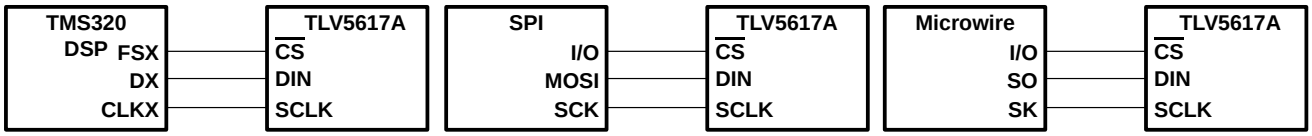



**Figure 12. Three-Wire Interface**

Notes on SPI and Microwire: Before the controller starts the data transfer, the software has to generate a falling edge on the pin connected to $\overline{CS}$. If the word width is 8 bits (SPI and Microwire) two write operations must be performed to program the TLV5617A. After the write operation(s), the holding registers or the control register of the DAC update automatically on the rising $\overline{CS}$ edge, ending the write cycle to the DAC. Note: After transfer of the LSB during a data or control write cycle, one additional rising edge on SCLK is required to reset the internal state machine. This edge can occur when $\overline{CS}$ is high or low, but must occur before the next falling $\overline{CS}$ edge that begins the following write cycle. Refer to the timing diagram for more information.

### serial clock frequency and update rate

The maximum serial clock frequency is given by:

$$f_{sclkmax} = \frac{1}{t_{whmin} + t_{wlmin}} = 20\ \mathrm{MHz}$$

The maximum update rate is:

$$f_{updatemax} = \frac{1}{16\left(t_{whmin} + t_{wlmin}\right)} = 1.25\ \mathrm{MHz}$$

Note that the maximum update rate is just a theoretical value for the serial interface, as the settling time of the TLV5617A should also be considered.

## APPLICATION INFORMATION

### data format

The 16-bit data word for the TLV5617A consists of two parts:

- Program bits (D15..D12)
- New data (D11..D0)

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| R1 | SPD | PWR | R0 | MSB | | | | 10 Data bits | | | | | LSB | 0 | 0 |

SPD: Speed control bit 1 → fast mode 0 → slow mode
PWR: Power control bit 1 → power down 0 → normal operation
On power up, SPD and PWD are reset to 0 (slow mode and normal operation)

The following table lists all possible combinations of register-select bits:

**register-select bits**

| R1 | R0 | REGISTER |
|---|---|---|
| 0 | 0 | Write data to DAC B and BUFFER |
| 0 | 1 | Write data to BUFFER |
| 1 | 0 | Write data to DAC A and update DAC B with BUFFER content |
| 1 | 1 | Reserved |

The meaning of the 12 data bits depends on the register. If one of the DAC registers or the BUFFER is selected, then the 12 data bits determine the new DAC value:

### examples of operation

- Set DAC A output, select fast mode:

  Write new DAC A value and update DAC A output:

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | New DAC A output value | | | | | | | | | | 0 | 0 |

The DAC A output is updated on the rising clock edge after D0 is sampled.

- Set DAC B output, select fast mode:

  Write new DAC B value to BUFFER and update DAC B output:

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | New BUFFER content and DAC B output value | | | | | | | | | | 0 | 0 |

The DAC A output is updated on the rising clock edge after D0 is sampled.

- Set DAC A value, set DAC B value, update both simultaneously, select slow mode:

  1. Write data for DAC B to BUFFER:

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | New DAC B value | | | | | | | | | | 0 | 0 |

  2. Write new DAC A value and update DAC A and B simultaneously:

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | New DAC A value | | | | | | | | | | 0 | 0 |

## APPLICATION INFORMATION

### examples of operation (continued)

Both outputs are updated on the rising clock edge after D0 from the DAC A data word is sampled.

- Set powerdown mode:

| D15 | D14 | D13 | D12 | D11 | D10 | D9 | D8 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| X | X | 1 | X | X | X | X | X | X | X | X | X | X | X | X | X |

X = Don't care

### linearity, offset, and gain error using single ended supplies

When an amplifier is operated from a single supply, the voltage offset can still be either positive or negative. With a positive offset, the output voltage changes on the first code change. With a negative offset, the output voltage may not change with the first code, depending on the magnitude of the offset voltage.

The output amplifier attempts to drive the output to a negative voltage. However, because the most negative supply rail is ground, the output cannot drive below ground and clamps the output at 0 V.

The output voltage then remains at zero until the input code value produces a sufficient positive output voltage to overcome the negative offset voltage, resulting in the transfer function shown in Figure 13.

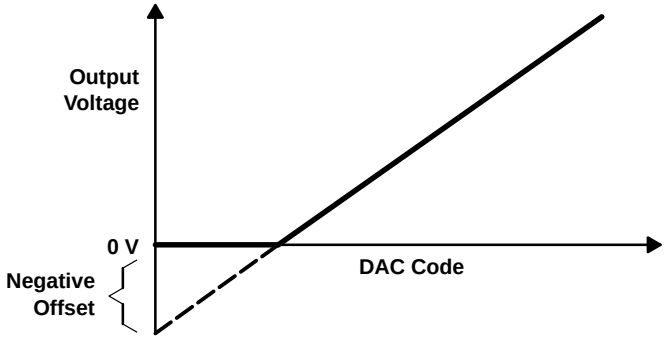



**Figure 13. Effect of Negative Offset (Single Supply)**

This offset error, not the linearity error, produces this breakpoint. The transfer function would have followed the dotted line if the output buffer could drive below the ground rail.

For a DAC, linearity is measured between zero-input code (all inputs 0) and full-scale code (all inputs 1) after offset and full scale are adjusted out or accounted for in some way. However, single supply operation does not allow for adjustment when the offset is negative due to the breakpoint in the transfer function. So the linearity is measured between full-scale code and the lowest code that produces a positive output voltage.

### definitions of specifications and terminology

#### integral nonlinearity (INL)

The relative accuracy or integral nonlinearity (INL), sometimes referred to as linearity error, is the maximum deviation of the output from the line between zero and full scale excluding the effects of zero code and full-scale errors.

#### differential nonlinearity (DNL)

The differential nonlinearity (DNL), sometimes referred to as differential error, is the difference between the measured and ideal 1 LSB amplitude change of any two adjacent codes. Monotonic means the output voltage changes in the same direction (or remains constant) as a change in the digital input code.

## definitions of specifications and terminology (continued)

**zero-scale error ($E_{ZS}$)**

Zero-scale error is defined as the deviation of the output from 0 V at a digital input value of 0.

**gain error ($E_G$)**

Gain error is the error in slope of the DAC transfer function.

**total harmonic distortion (THD)**

THD is the ratio of the rms value of the first six harmonic components to the value of the fundamental signal. The value for THD is expressed in decibels.

**signal-to-noise ratio + distortion (S/N+D)**

S/N+D is the ratio of the rms value of the output signal to the rms sum of all other spectral components below the Nyquist frequency, including harmonics but excluding dc. The value for S/N+D is expressed in decibels.

**spurious free dynamic range (SFDR)**

Spurious free dynamic range is the difference between the rms value of the output signal and the rms value of the largest spurious signal within a specified bandwidth. The value for SFDR is expressed in decibels.

## PACKAGING INFORMATION

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| TLV5617ACD | Active | Production | SOIC (D) \| 8 | 75 \| TUBE | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | TV5617 |
| TLV5617ACD.A | Active | Production | SOIC (D) \| 8 | 75 \| TUBE | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | TV5617 |
| TLV5617ACDG4 | Active | Production | SOIC (D) \| 8 | 75 \| TUBE | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | TV5617 |
| TLV5617ACDR | Active | Production | SOIC (D) \| 8 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | TV5617 |
| TLV5617ACDR.A | Active | Production | SOIC (D) \| 8 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | TV5617 |
| TLV5617AID | Active | Production | SOIC (D) \| 8 | 75 \| TUBE | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | TY5617 |
| TLV5617AID.A | Active | Production | SOIC (D) \| 8 | 75 \| TUBE | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | TY5617 |
| TLV5617AIDR | Active | Production | SOIC (D) \| 8 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | TY5617 |
| TLV5617AIDR.A | Active | Production | SOIC (D) \| 8 | 2500 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | -40 to 85 | TY5617 |

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

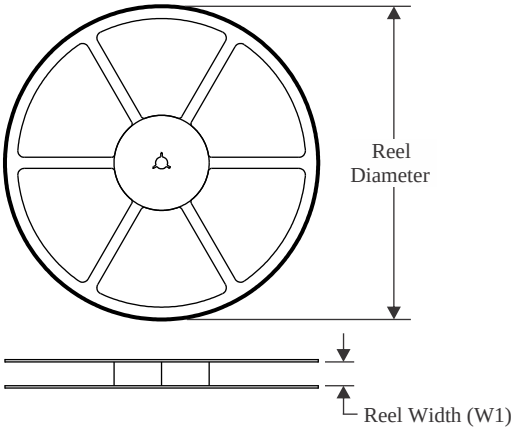
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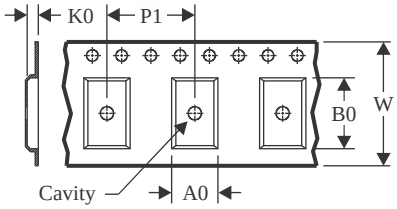
In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

## TAPE AND REEL INFORMATION

**REEL DIMENSIONS**

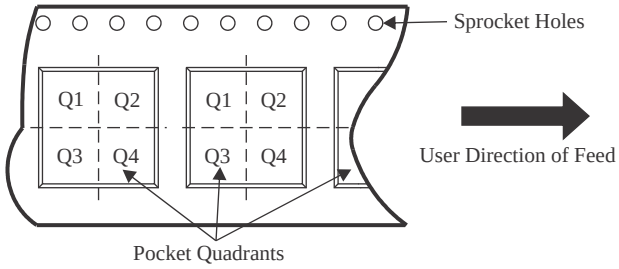



**TAPE DIMENSIONS**




| | |
|---|---|
| A0 | Dimension designed to accommodate the component width |
| B0 | Dimension designed to accommodate the component length |
| K0 | Dimension designed to accommodate the component thickness |
| W | Overall width of the carrier tape |
| P1 | Pitch between successive cavity centers |

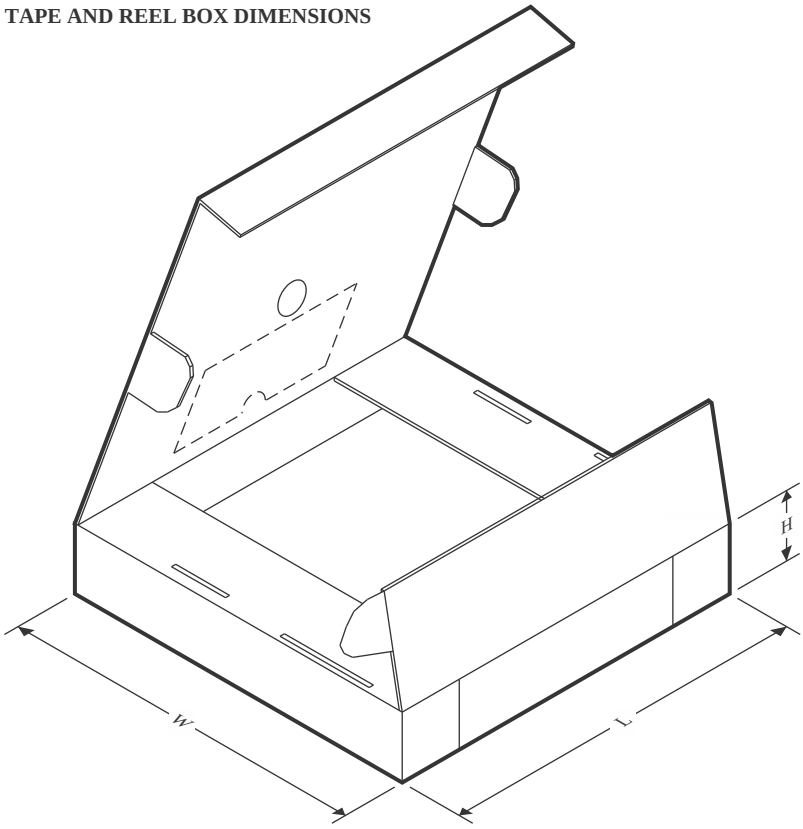
**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| TLV5617AIDR | SOIC | D | 8 | 2500 | 330.0 | 12.4 | 6.4 | 5.2 | 2.1 | 8.0 | 12.0 | Q1 |

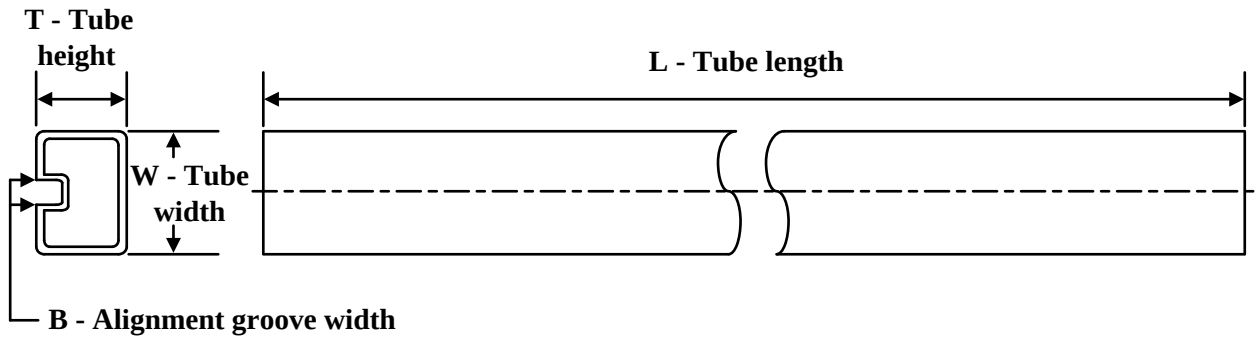
TAPE AND REEL BOX DIMENSIONS




*All dimensions are nominal

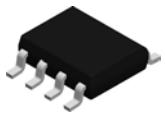
| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| TLV5617AIDR | SOIC | D | 8 | 2500 | 350.0 | 350.0 | 43.0 |

## TUBE




*All dimensions are nominal

| Device | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (µm) | B (mm) |
|---|---|---|---|---|---|---|---|---|
| TLV5617ACD | D | SOIC | 8 | 75 | 505.46 | 6.76 | 3810 | 4 |
| TLV5617ACD.A | D | SOIC | 8 | 75 | 505.46 | 6.76 | 3810 | 4 |
| TLV5617ACDG4 | D | SOIC | 8 | 75 | 505.46 | 6.76 | 3810 | 4 |
| TLV5617AID | D | SOIC | 8 | 75 | 505.46 | 6.76 | 3810 | 4 |
| TLV5617AID.A | D | SOIC | 8 | 75 | 505.46 | 6.76 | 3810 | 4 |

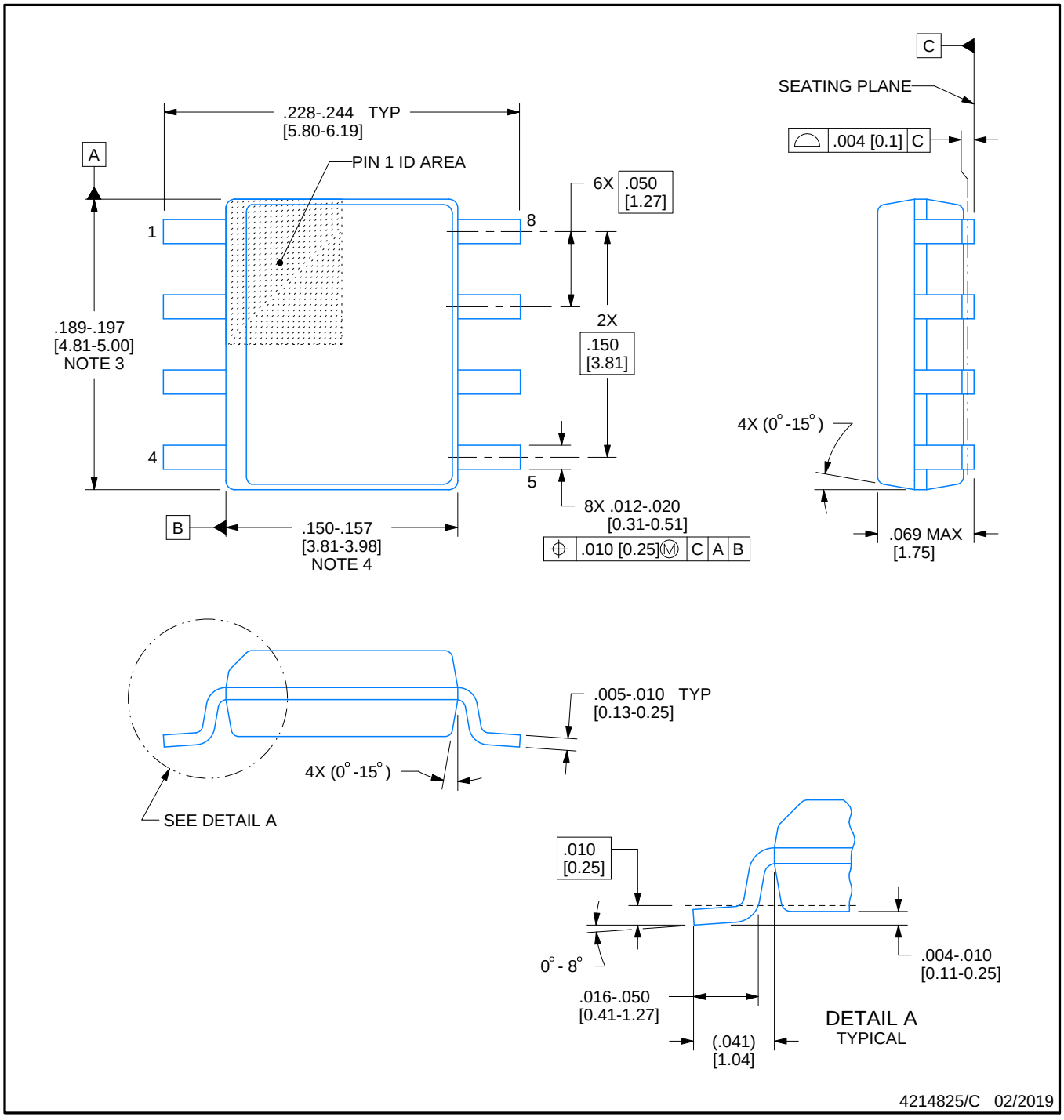


# PACKAGE OUTLINE

## D0008A

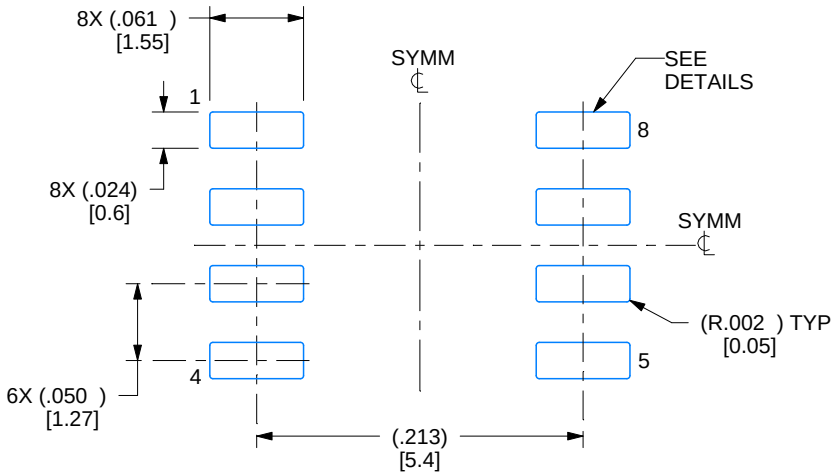
## SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT

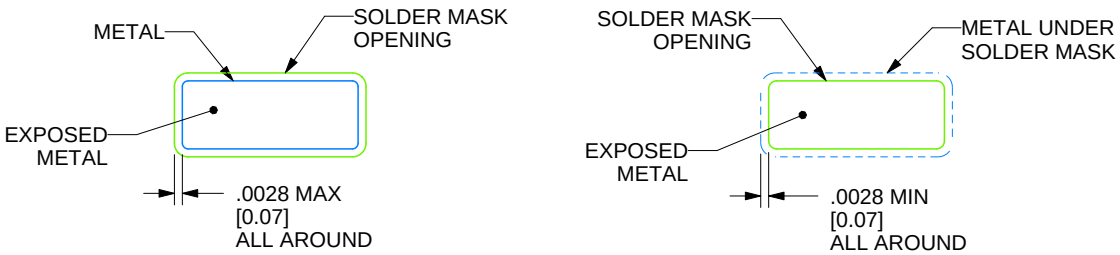



NOTES:

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.




LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X




SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

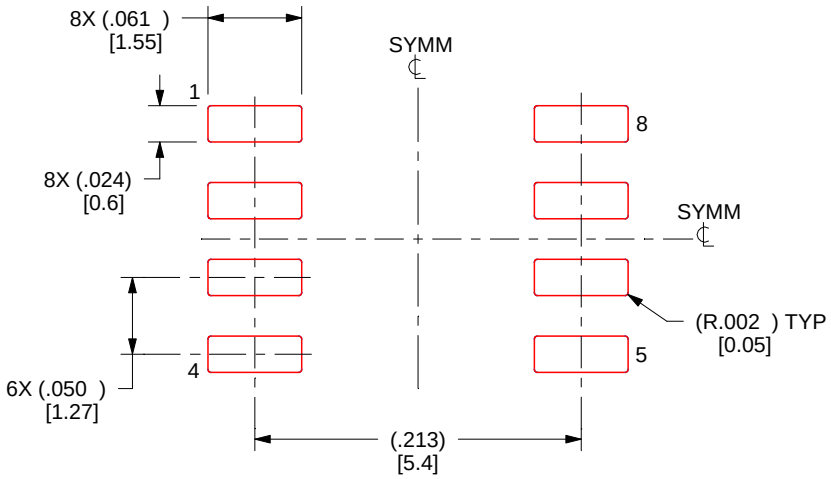
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

## D0008A

## SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT




SOLDER PASTE EXAMPLE
BASED ON .005 INCH [0.125 MM] THICK STENCIL
SCALE:8X

4214825/C 02/2019

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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